



CS 510 • Advanced Information Retrieval • Spring 2026

Beyond Single-Answer RAG: Viewpoint Coverage Retrieval for Financial Decision Support

RESEARCH PROJECT PRESENTATION

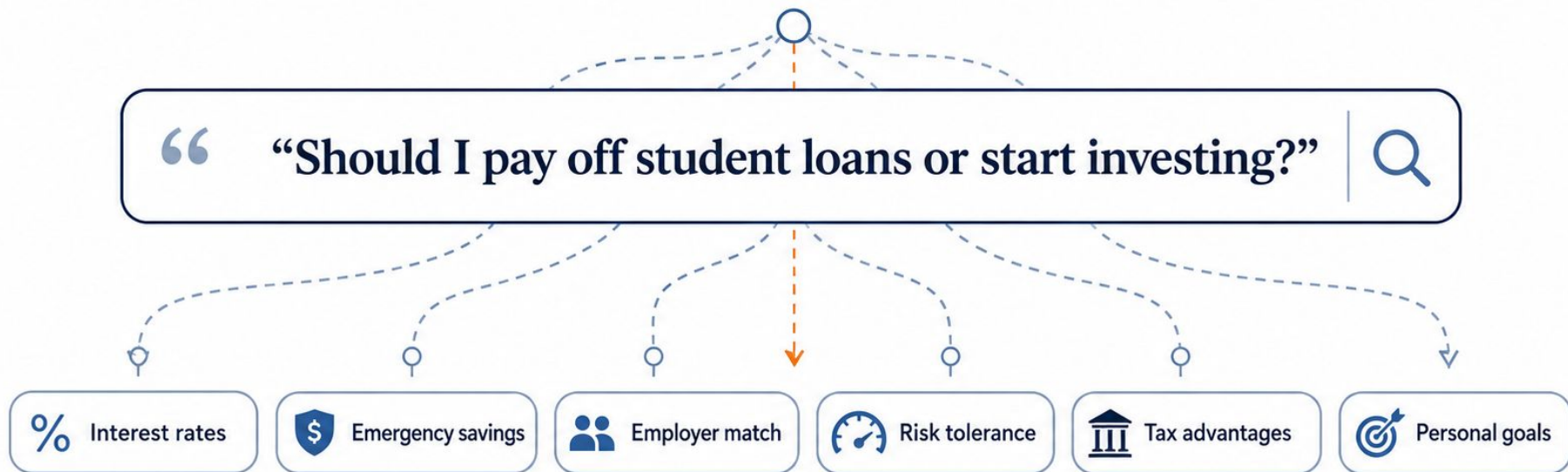


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Decision-Oriented Queries Are Different



— • **Not one answer — a decision landscape.** • —



Why Existing Systems Fall Short



1. Why not normal search?

- Returns ranked links, not organized viewpoints
- Dominant framings can crowd out alternatives
- Users must manually compare evidence and caveats



Good for finding sources,
weak for deliberation



2. Why not LLMs?

- May produce unsupported or hallucinated claims
- Usually provides no explicit grounding to evidence
- Compresses multiple viewpoints into one answer



Fluent, but not
reliably grounded



3. Why not RAG?

- Improves grounding by retrieving evidence
- But still synthesizes a single narrative
- Disagreement, caveats, and minority views may be flattened



Grounded, but still
viewpoint-compressed

» Need: grounded multi-perspective retrieval, **not a single final answer.**



From Search to EPIC Viewpoint Coverage Retrieval

Research Question and Task



Research Question

Can a retrieval system organize decision-oriented financial advice into coherent, distinct, and grounded perspectives while preserving conditions and caveats?



Introducing EPIC

EPIC = Epistemic Progressive Interest Clustering

Our pipeline for viewpoint coverage retrieval.



major decision
directions



supporting
reasons



applicability
conditions



caveats and
uncertainty



source
evidence



What Changes?

Not:

- one ranked list
- one synthesized answer

Instead:

- multiple grounded perspectives
- reasons, conditions, and caveats
- no forced final verdict



EPIC's goal: retrieve the decision landscape — **not** a final verdict.



Motivation and Related Work

Where EPIC Fits



EPIC builds on prior work in search diversification, argument retrieval, RAG, diverse/viewpoint-oriented RAG, and HCI overreliance — and extends them by **organizing grounded viewpoints** instead of producing a single final answer.

Prior Work Area	What It Shows	Key References	Gap EPIC Addresses
 Search Diversification	Reduces redundant results and improves semantic variety.	Carbonell & Goldstein (1998); Santos et al. (2010)	Diversifies documents, not grounded viewpoints, conditions, or caveats.
 Argument Retrieval	Retrieves pro/con arguments on controversial topics.	Bondarenko et al. (2020)	Often binary; financial advice is conditional, hybrid, and multi-perspective.
 Standard RAG	Grounds LLM responses in retrieved documents.	Lewis et al. (2020)	Improves grounding, but still synthesizes a single answer and inherits corpus skew.
 Diverse / Viewpoint-Oriented RAG	Seeks multiple answers or viewpoints instead of only one response.	DF-RAG; DIVERGE	Usually remains generation-centered; EPIC builds an auditable perspective map directly from retrieved evidence.
 HCI Overreliance Work	Users may over-trust AI and rely on it passively.	Buccinca et al. (2021)	Focuses on interface interventions, not retrieval design or auditable viewpoint coverage.



Our Direction

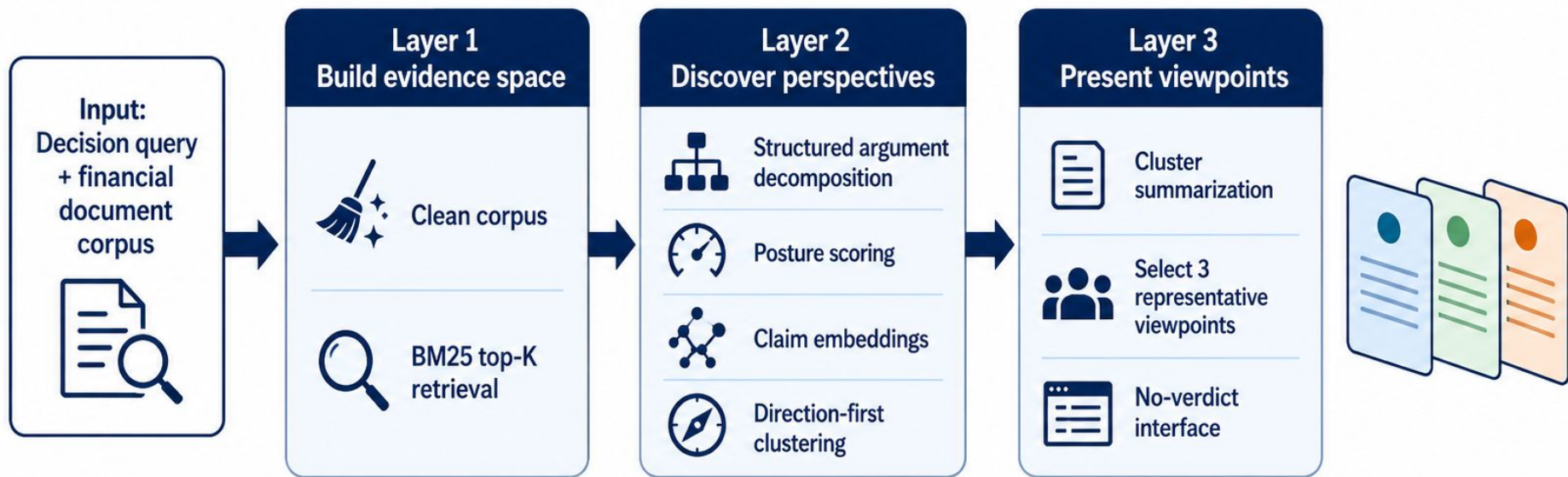
EPIC shifts from diverse answer generation to viewpoint coverage retrieval: retrieve evidence, decompose arguments, organize grounded perspectives, and present reasons, conditions, and caveats without forcing a final recommendation.

Goal:

Grounded, auditable perspective maps for deliberative decision-making.



How EPIC Turns Documents into Perspective Panels



EPIC is a **retrieval-to-perspective pipeline** — not just a clustering algorithm.



Core Design Choices Behind EPIC



1. Evidence-first retrieval

BM25 top-K builds the evidence pool before organization.

Why: retrieval depth determines which viewpoints are even visible.



2. Decompose before clustering

Extract main claim, reasons, conditions, caveats, and direction.

Initial failure: full passages mixed claims with caveats, so clusters collapsed into generic 'it depends' groups.



3. Cluster claims, not passages

Use main-claim embeddings for viewpoint discovery.

Why: the recommendation should drive clustering; conditions stay available for display.



4. Direction-first discovery

Separate option A, option B, and hybrid before semantic sub-clustering.

Why: semantically similar finance language can hide opposing recommendations.



Takeaway: EPIC organizes recommendations, preserves nuance, and avoids collapsing all advice into one answer.



EVALUATION DESIGN

Dataset & Ablation Baselines



5 Financial Decision Queries

- Q1** Student loans vs. investing
- Q2** Roth IRA vs. traditional 401(k)
- Q3** Emergency fund vs. credit-card debt
- Q4** Buy vs. rent
- Q5** Index funds vs. mortgage payoff



Retrieval depths: $K = 5, 10, 15, 20$

Ablation Baselines



Random
lower-bound sanity check



RelevanceOrder
tests BM25 rank alone



SemanticOnlyGlobal
strongest intrinsic baseline



DirectionOnly
direction labels without semantic refinement



NoUnclearHandling
tests vague/unclear argument handling



Takeaway: We test whether EPIC adds value beyond retrieval order, semantic clustering, and direction labels alone.



Geometric & Epistemic Metrics

1. Are clusters coherent?



Purity

within-cluster similarity:
do arguments in the same
panel say the same thing?



InterDist

between-cluster distance:
are perspective panels
actually different?



Good clusters should be
internally tight and mutually distinct.

2. ECS: coherence × distinctiveness

$$ECS = \sum_c Purity(c) \times \log \left(\frac{K}{\text{mean_sim}(c, \text{others})} \right)$$

TF-IDF:

term frequency ×
rarity across
documents

VS

ECS:

cluster purity ×
rarity among
other clusters



Rewards clusters that are both
focused and ***non-redundant***.

3. Balance-adjusted ECS

$$BalAdj = ECS \times \text{normalized entropy}$$

Primary metric: penalizes skewed
perspective maps.

Bad:

10 debt-first,
0 hybrid



high purity, poor coverage

Good:

debt-first +
invest-first +
hybrid



balanced deliberation



CovAdj

CovAdj = category coverage



BalAdj

BalAdj = proportional balance



DirBal

DirBal = direction entropy



PostBal

PostBal = posture entropy



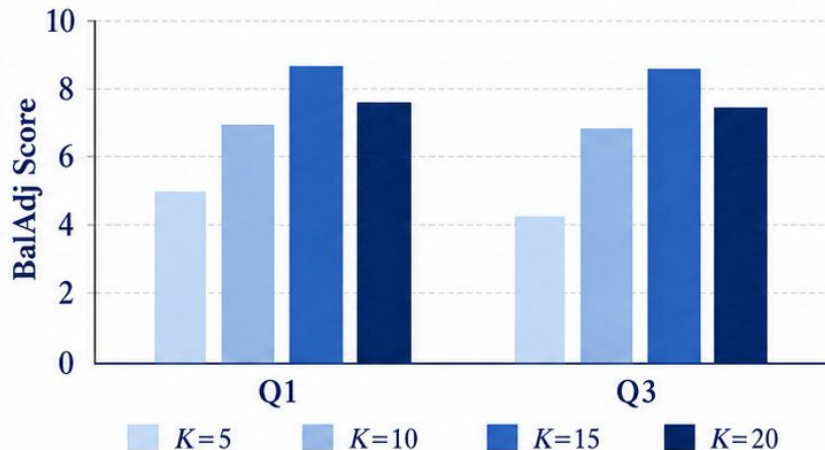
Takeaway: We test not only whether clusters are clean, but whether they preserve the range of decision-relevant viewpoints.



KEY RESULTS

Retrieval Depth as an Epistemic Control Knob

BalAdj scores rise with useful evidence, then plateau



Illustrative BalAdj pattern for Q1 and Q3: K=15 is the strongest general-purpose setting.



Shallow pools ($K=5$)

Too few extracted arguments; major viewpoint families can be missed.



Sweet spot ($K=15$)

Best balance between viewpoint coverage and redundancy; strongest for Q1, Q3, and Q5.



Deep pools ($K=20$)

Often adds same-side variants faster than new user-facing perspectives.



Graceful degradation

When evidence is thin, EPIC stays narrow and grounded instead of inventing unsupported views.



Takeaway: In viewpoint coverage retrieval, K is not just an implementation setting — it controls which perspectives are available for discovery.



KEY RESULTS

EPIC vs. Semantic-Only Baselines

Same evidence, different organization

EPIC

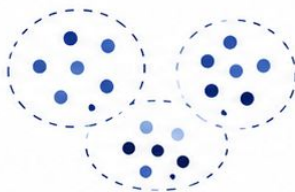
- Direction-aware scaffolding
- Decomposes claims from caveats
- Preserves stance families
- Deliberation-ready perspective panels



VS

SemanticOnlyGlobal

- Embedding-only clustering
- Strong intrinsic baseline
- Competitive on raw geometry
- Can merge semantically similar but opposing recommendations



SemanticOnlyGlobal is a strong baseline — not a straw man.

Where EPIC dominates



Q1 (Student loans) and Q3 (Emergency fund) show naturally separable stance families. EPIC's direction-first scaffolding yields clearer debt-first, invest-first, and hybrid panels.

Where semantics hold their ground



On noisier or structurally asymmetric corpora such as Q4 (Buy vs. Rent), SemanticOnlyGlobal remains highly competitive. This confirms that semantic clustering is a strong intrinsic baseline.

Human-centered advantage



Even when raw metrics are close, EPIC's preservation of distinct stances, conditions, and caveats creates a more interpretable, deliberation-ready UI.



Takeaway: EPIC's contribution is not just cleaner clustering, but a more faithful and usable organization of competing viewpoints.



Conclusion, Limitations & Future Work



1. Conclusion

EPIC reframes RAG from single-answer synthesis to viewpoint coverage retrieval.

Works best when the corpus contains separable stance families.

Key contribution: deliberation-facing organization of grounded perspectives.



2. Limitations

Small corpora; retrieval depth and corpus quality still constrain coverage.

Decomposition and summarization rely on LLM prompting.

Evaluation is mostly intrinsic; no user study yet.



3. Future Work

Scale to larger and more diverse financial corpora.

Add human evaluation of coherence, usefulness, and faithfulness.

Compare EPIC against search and single-answer RAG on trust calibration and decision quality.



Takeaway: Decision-support RAG should preserve disagreement, assumptions, and caveats — not compress evidence into one answer.